Pet Management App

Project Introduction and Core Modules

Prepared for the SE2020 - Web and Mobile Technology Project

Pet Management App is a comprehensive mobile application designed to simplify pet ownership by bringing pet profile management, pet health tracking, service booking, social interaction, user feedback, and AI-powered assistance into one platform. The purpose of this application is to help users manage their pets' daily needs in a smart, organized, and user-friendly way. The system also includes both user and admin management features to support smooth operation and better service delivery.

Project Introduction

This project is developed as a full-stack mobile application focused on pet care and pet-related activities. It provides a single digital space where users can maintain pet information, record health details, book pet services, share social updates, and submit feedback. The app is designed to improve convenience for pet owners while also allowing administrators to manage users and services effectively.

The application follows a CRUD-based architecture, where each module is responsible for creating, reading, updating, and deleting data. This structure makes the system organized, scalable, and easier to maintain. In addition to the main CRUD modules, the system also integrates AI assistance to provide helpful pet-care guidance and smart support for users.

Core CRUD Modules

1. User Management CRUD (User + Admin) - Default

The User Management CRUD module is the default system module that handles account-related operations for both normal users and administrators.

What this module does

- Create new user and admin accounts
- View user profile details
- Update account information
- Delete user accounts when necessary

Main data handled

- Name
- Email
- Password

- Phone number
- Address
- Profile image
- Role type (User/Admin)

Purpose: This module serves as the foundation for authentication, authorization, and access control within the system.

2. Pet Management CRUD - Komal

The Pet Management CRUD module allows users to create and manage profiles for their pets.

What this module does

- Add a new pet profile
- View pet details
- Update pet information
- Delete a pet profile

Main data handled

- Pet name
- Pet type
- Breed
- Age
- Gender
- Weight
- Color
- Pet image

Purpose: This module maintains complete and organized details about each pet and acts as one of the main core modules of the application.

3. Pet Health Management CRUD - Nehara

The Pet Health Management CRUD module manages all important health-related information of pets in one place. It combines medical records, vaccination details, and growth or weight tracking into a single module.

What this module does

- Add health-related records
- View pet health history
- Update medical, vaccination, and growth details
- Delete unwanted or incorrect records

Main data handled

- Diagnosis
- Treatment details
- Medicines

- Vaccination name
- Vaccination date
- Next vaccination date
- Weight records
- Growth notes
- Vet notes
- Check-up date

Purpose: This module helps users manage the complete health history of their pets in a simple and organized way.

4. Service Booking CRUD - Shymal

The Service Booking CRUD module allows users to book pet-related services through the application.

What this module does

- Create a service booking
- View booked services
- Update or reschedule bookings
- Cancel or delete bookings

Main data handled

- Pet ID
- Service type
- Booking date
- Booking time
- Notes
- Booking status

Purpose: This module helps users access pet-related services such as grooming, boarding, walking, or training in a convenient way.

5. Service Management CRUD (Admin) - Monishka

The Service Management CRUD module is handled by the admin and is used to manage the services available in the application.

What this module does

- Add new services
- View available services
- Update service details
- Delete services

Main data handled

- Service name
- Service category

- Description
- Price
- Availability
- Service image

Purpose: This module allows the admin to maintain the list of services offered through the application so that users can browse and book them.

6. Social Feed CRUD - Savini

The Social Feed CRUD module allows users to share pet-related posts and interact with the community.

What this module does

- Create posts
- View posts
- Update their own posts
- Delete their own posts

Main data handled

- Post caption
- Post image or media
- Post date
- Pet reference
- Likes count
- Comments count

Purpose: This module creates a social environment where pet owners can share updates, photos, and experiences related to their pets.

7. Feedback CRUD - Wathsala

The Feedback CRUD module allows users to submit and manage feedback related to services, app features, or their overall experience.

What this module does

- Create feedback
- View submitted feedback
- Update feedback
- Delete feedback

Main data handled

- Feedback title
- Feedback message
- Rating
- Date
- Related service or feature

Purpose: This module collects user opinions and experiences in order to improve the quality and usability of the application.

Tools and Technology

The Pet Management App uses modern web and mobile development tools to build a scalable, secure, and user-friendly full-stack solution.

Layer	Technology	Purpose
Frontend (Mobile)	React Native (Expo)	Builds the cross-platform mobile application interface.
Backend	Node.js with Express.js	Handles REST API development, business logic, and request processing.
Database	MongoDB	Stores user accounts, pet records, bookings, services, posts, and feedback data.
Authentication	JWT / Firebase Auth	Supports secure login, registration, and session handling.
Image Storage	Cloudinary	Stores pet images, service images, and social feed media efficiently.
Notifications	Expo Notifications / Firebase Cloud Messaging	Used for reminders and important user updates.
Development Tools	VS Code, Postman, GitHub	Used for coding, API testing, and version control.

AI Assistance

AI Assistance is included as an advanced supporting feature of the application. It is designed to provide smart pet-care guidance and improve the overall user experience.

Through this feature, users can ask general pet-care questions, receive feeding and care suggestions, obtain basic health guidance, and get helpful tips based on common pet needs. This feature is intended only for general assistance and does not replace professional veterinary advice.